

ASM-I: Week 01 Notes

Non-Parametric Methods

Introduction to Hypothesis Tests

Non-Parametric (Distribution-free methods for testing hypothesis)

Recap: Parametric Methods

1. One Sample Mean

- Assumed Model: $E[X_i] = \mu$
- Hypothesis Tests:
 - $H_0 : \mu = \mu_0$
 - $H_a : \mu \neq \mu_0$

- Test Statistic:

$$T = \frac{\bar{X} - \mu_0}{S/\sqrt{n}}$$

- Assumptions:

- $X_1, \dots, X_n \stackrel{iid}{\sim} F$
- If n is large, F can be any distribution (CLT applies).
- If n is small, we assume $F \sim N$ (Normal Distribution).
- What if n is small and F is not Normal?

2. Two Sample Mean

- Samples: $X_1, \dots, X_n \sim F$ and $Y_1, \dots, Y_m \sim G$
- Assumption: $E[X] = \mu_1, E[Y] = \mu_2$
- Hypothesis:
 - $H_0 : \mu_1 = \mu_2$
 - $H_a : \mu_1 \neq \mu_2$
- Conditions:

- If F and G are both Normal $\rightarrow$ **2-sample t-test**.
- If both m and n are large $\rightarrow$ **Normal test**.

3. Variance Test (One Sample)

- Sample: $X_1, \dots, X_n \sim F$
- Variance: $\text{Var}(X_i) = \sigma^2$
- Statistic: $\frac{(n-1)S^2}{\sigma_0^2} \sim \chi_{n-1}^2$
- S^2 is unbiased sample variance
- Hypothesis: $H_a : \sigma^2 \neq \sigma_0^2$
- **Assumption:** F is Normal.

Comparing Parametric vs. Non-Parametric

4. Two Sample Variance

- Statistic: $\frac{S_1^2}{S_2^2}$
- Hypothesis: $H_0 : \sigma_1^2 = \sigma_2^2$
- Test: **F-test**
- Assumption: $F = N = G$ (Both distributions are Normal).

5. ANOVA (One-way)

- Equality of means of several samples.
- Valid under Normal assumptions.

Motivation for Non-Parametric Tests

What if the sample size (n) is small and the parent distribution is unknown?

- If the parent distribution is known but *not* Normal, we may be able to construct the distribution of the test statistic (e.g., if $F \sim \text{Exp}(\lambda)$, $\sum X_i \sim \text{Gamma}(n, \lambda)$. $F \sim \text{Pois}(\lambda)$, $\sum X_i \sim \text{Pois}(n\lambda)$) Then we can construct the test statistic under H_0 using the knowledge of probability theory.
- However, often we do not know the distribution.

Definition: The tests we shall construct are **Distribution Free**.

- We do not assume any parametric model (a model with known form but finite number of unknown parameters).

Why not always use Non-Parametric methods?

- The flexibility is attractive but comes with a cost.

- When a parametric model is true/correct, then using that parametric test is “better”.
- **Power:** For hypothesis testing, both parametric and non-parametric tests will have the same significance level α . But the parametric test (under the correct parent model) will have **more power**.
- **Confidence Intervals:** Both procedures will give the same average probability (coverage), but the parametric interval will be **shorter/narrower** (more precise).

The Sign Test (One Sample Median Test)

Use non-parametric techniques if the distribution of the data is unknown.

One Sample Median Test

(Alternative to one sample t-test)

Concept:

- Generally for F , mean may not exist (e.g., Cauchy), but **Median always exists**.
- Median equals Mean if $F \approx N$ (Normal).
- Definition: Median $M = \min\{a : F(a) \geq 1/2\}$.

Example Scenario: Arsenic Concentration

- Sources statlib (CMU) Arsenic in drinking water
- Data: $X_1, \dots, X_{21}$ (Arsenic in toenail clippings of 21 individuals).
- Hypothesis: $H_0 : \theta = 0.2$ vs $H_a : \theta > 0.2$ (Right Tailed).
- Let Median(X_i) = θ .

Procedure: 1. Under H_0 , observe $P(X_i > 0.2) = 1/2$.

2. Sample $X_1, \dots, X_n \stackrel{iid}{\sim} F$ (Independent).
3. Define Indicator Variable Y_i :

$$Y_i = \begin{cases} 1 & \text{if } X_i > \theta_0 \\ 0 & \text{if } X_i \leq \theta_0 \end{cases}$$

4. Test Statistic:

$$Y = \sum_{i=1}^n Y_i \sim \text{Bin}\left(n, \frac{1}{2}\right)$$

(Since X_i are independent, Y_i are independent Bernoulli trials).

Rejection Region:

- Reject for large values of Y .
- $P(Y > C) = \alpha$ (Right Tailed).
- Depending on the type of H_a , you may need a Left Tailed or Two Tailed rejection region.

Sign Test Analysis & Wilcoxon Intro

Randomness Check: Randomness test is necessary.

Example Calculation:

- $H_a : \theta > 0.2$ and observed $y = 10$ with $n = 21$.
- P-value calculation (using R): 2. `pbinom(10, 21, 0.5)=1.`
- If p-value $> \alpha$, do not reject H_0 .

Large Sample Approximation:

- If n is large, use Normal approximation to Binomial (converges to normal quickly $n \geq 10$) and not to general F (may take long time, much larger n needed).
- Test Statistic $Y \sim \text{Bin}(n, 0.5)$.
- $n = 21, y = 10$
- $Z = \frac{Y - np}{\sqrt{npq}} = \frac{Y - 0.5n}{0.5\sqrt{n}}.$
- 2. `pnorm(Z)=0.8`
- Do not reject H_0 at any $\alpha < 0.8$.

Summary of Sign Test:

- It is a one-sample median test looking at the sign of $X_i - \theta_0$.
 - Widely applicable. The only assumption is that F is **continuous**.
 - **Drawback:** Low power under many alternatives. It ignores the *magnitude* of the difference.
 - Example: A density could be symmetric but shaped such that signs are balanced even if the bulk of data is far off.
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Wilcoxon Signed Rank Test

- Assumes F is **continuous** and **symmetric**.
- Ranks are distribution free.
- **Idea:** Give more weight to X_i that are much larger than θ_0 (large deviation).

Procedure:

1. Calculate absolute differences: $|X_i - \theta_0|$.
2. Assign Ranks R_i to these absolute differences.
3. Max Sum of Ranks: $\sum R_i = \frac{n(n+1)}{2}$.
4. Test Statistic:

$$T = \sum R_i Y_i$$

(Where Y_i is the indicator of positive sign, as defined before).

$$Y_i = \begin{cases} 1 & \text{if } X_i > \theta_0 \\ 0 & \text{if } X_i \leq \theta_0 \end{cases}$$

Wilcoxon Signed Rank Example

Example: Heart Rate

- H_0 : Median = 70 vs H_a : Median > 70.
- $\theta_0 = 70$.
- Data (X_i): 73, 82, 87, 68, 106, 60, 97.

Calculation Table:

X_i	$X_i - \theta_0$	Sign (Y_i)	$ X_i - \theta_0 $	Rank (R_i)
73	+3	1	3	2
82	+12	1	12	4
87	+17	1	17	5
68	-2	0	2	1
106	+36	1	36	7
60	-10	0	10	3
97	+27	1	27	6

Calculate Statistic T:

$$T = \sum R_i Y_i = 2 + 4 + 5 + 7 + 6 = 24$$

Distribution of T

- Let $Y = \sum_{i=1}^n Y_i$ (Number of positive signs).
- $T = \sum R_i Y_i$ is the sum of ranks corresponding to positive values of $X_i - \theta_0$.
- Let $r_1 < \dots < r_y$ be the ranks corresponding to positive differences vector of signs $(r_1, \dots, r_n)$.
- $Y \sim \text{Bin}(n, 0.5)$
- Under H_0 , the vector of signs $(Y_1, \dots, Y_n)$ can take any of the 2^n values (e.g., 00...0 to 11...1) with equal probability $\frac{1}{2^n}$.

Exact Null Distribution

Each configuration of signs gives a value for T . We count how many configurations sum to K to find $P(T = K)$.

Example: $n = 7$, Find $P(T = 9)$ We need to find subsets of ranks $\{1, 2, 3, 4, 5, 6, 7\}$ that sum to 9.

- Length 2: (2, 7), (3, 6), (4, 5)
- Length 3: (1, 2, 6), (1, 3, 5), (2, 3, 4)

Total combinations = 6.

$$P(T = 9) = \frac{6}{2^7}$$

Other Probabilities ($n = 7$):

- $T = 0$: Empty set (all signs 0). $P(T = 0) = \frac{1}{2^n}$.
- $T = 1$: Set $\{1\}$. $P(T = 1) = \frac{1}{2^n}$.
- $T = 2$: Set $\{2\}$. $P(T = 2) = \frac{1}{2^n}$.
- $T = 3$: Sets $\{3\}, \{1, 2\}$. $P(T = 3) = \frac{2}{2^n}$.

The density is to evaluate the test statistic and find the p-value.

Comparison of Tests

1. **t-test:** Assumes $F \equiv N$ (Normal).
2. **Sign test:** Assumes F is continuous.
3. **Signed Rank test:** Assumes F is symmetric and continuous.

Performance:

- If the assumption of symmetry holds, the **Signed Rank test is more powerful** than the Sign test.
- If the assumption of symmetry does **not** hold, the p-value you get from the Signed Rank test will be **wrong**.

How do the tests compare

$$\text{one sample test for centre} \left\{ \begin{array}{ll} \text{t-test} & F \equiv N \\ \text{Sign test} & F \text{ is continuous} \\ \text{Signed Rank test} & F \text{ is symmetric \& continuous.} \end{array} \right. \quad \text{same level}$$

Large Sample Approximation

Distribution Parameters for T:

- Mean: $E[T] = \frac{n(n+1)}{4}$
- Variance: $\text{Var}(T) = \frac{n(n+1)(2n+1)}{24}$

Standardized Statistic:

$$T^* = \frac{T - E[T]}{\sqrt{\text{Var}(T)}} \approx N(0, 1)$$

Derivation: Let $U_i = R_i Y_i$. Here ranks are fixed numbers $1, \dots, n$ but Y_i is random. However, we can view this as:

$$T = \sum_{i=1}^n U_i$$

$$V_i = \begin{cases} i & \text{with probability } \frac{1}{2} \quad (\text{if sign is positive}) \\ 0 & \text{with probability } \frac{1}{2} \quad (\text{if sign is negative}) \end{cases}$$

Abstract

We define V_i as the contribution of the i -th rank to the total sum T . Instead of summing randomized ranks, we look at the fixed rank value i (e.g., rank 1, rank 2, rank 5) and ask:

Will this rank be added to our total? If the sign is positive: We add the rank value i to our total. (Value = i).

If the sign is negative: We do not add anything. (Value = 0).

Since the sign is a "coin flip" (50/50 chance) under H_0 : V_i takes the value i with probability $1/2$. V_i takes the value 0 with probability $1/2$

- $E[V_i] = i \cdot \frac{1}{2} + 0 \cdot \frac{1}{2} = \frac{i}{2}$
- $E[T] = \sum_{i=1}^n \frac{i}{2} = \frac{1}{2} \frac{n(n+1)}{2} = \frac{n(n+1)}{4}$
- $\text{Var}(V_i) = E[V_i^2] - (E[V_i])^2 = \frac{i^2}{2} - \frac{i^2}{4} = \frac{i^2}{4}$
- $\text{Var}(T) = \sum_{i=1}^n \frac{i^2}{4} = \frac{1}{4} \sum i^2 = \frac{1}{4} \frac{n(n+1)(2n+1)}{6} = \frac{n(n+1)(2n+1)}{24}$

Note:

- V_i are independent as Y_i are independent.
- This approximation is useful when n is moderately large.
- Computation of 2^n possibilities is very time consuming for large n .

Handling Ties

What happens if data contains ties?

Case 1: Zeros

- If $X_i - \theta_0 = 0$ for same i .
- **Method:** Drop those observations and reduce n .

Case 2: Equal Absolute Differences

- If $|X_i - \theta_0| = |X_j - \theta_0|$ for $i \neq j$
- **Method:** Give them the **Average Rank**.

Example: Values to rank: 5, 6, 5, 6, 7 1. Sort them: 5, 5, 6, 6, 7 2. Initial positions: 1, 2, 3, 4, 5 3. Average ties:

- The two 5s are at positions 1 and 2. Average rank = $\frac{1+2}{2} = 1.5$.
- The two 6s are at positions 3 and 4. Average rank = $\frac{3+4}{2} = 3.5$.

- The 7 is at position 5. Rank = 5.
4. Final Ranks: 1.5, 1.5, 3.5, 3.5, 5.
- Note:** When ties are present and average ranks are used, the p-value derived is **approximate**, not exact.

```
Week_I
#This generates all possible subsets of size 2 when $n=4$.
combn(c(1:4),2)

#Writing own function for p-value of signed rank test
signrank<-function(n,T){
  x<-c(1:n)
  k<-0
  for(i in 1:n){
    y<-combn(x,i)
    l<-dim(y)[2]
    for(j in 1:l){
      k<-cbind(k,sum(x[y[,j]]))}
    }
  pval<-length(which(k>=T))/length(k)
  return(pval)
}
table(k)
#Signed rank test for the example done in class
a<-c(73, 82, 87, 68, 106, 60,97)
b<-a-70
r<-rank(abs(b))
s<-ifelse(b>0,1,0)
T<-sum(r*s)
signrank(7,T)

# Directly using built-in function for signed rank test
library(stats)
wilcox.test(a,mu=70,alternative="greater")
```